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Net Zero by 2050: investigating carbon-budget compliant retrofit measures for the English housing stock

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Abstract

The UK has enacted one of the most ambitious carbon reduction targets striving for net zero emissions by 2050. A major challenge to achieving this is decarbonizing heating demand with almost 25m homes in need of retrofit. This paper explores a range of retrofit interventions to the English residential stock, and their impact on embodied and operational carbon emissions from 2021-2050; comparing these to potential national carbon budgets. The results show that mass deployment of air source or ground source heat pumps can reliably achieve combined embodied and operational emissions within the national carbon budget by 2050. Whilst the careful selection of insulation materials is key in bringing down the embodied emissions particularly for meeting more strict carbon budgets. These are, however, under the assumption that all retrofit combinations are deployed immediately at the start of 2021. The identified scale and pace of deployment thus needed to stay within carbon budgets is likely to pose enormous practical challenges. To overcome these practical challenges, we argue for (a) increasing both heat pump deployment and renewable generation targets beyond existing pledges, (b) increasing awareness of residential retrofit benefits, and (c) providing more attractive financing options to incentivise and increase retrofit uptake.

Keywords Retrofit scenarios, Decarbonisation pathways, Embodied carbon, Whole life carbon, Residential stock

1 Introduction

To keep the global temperature rise to within 1.5°C in line with the Paris climate accord, drastic decarbonisation measures are needed (Morton, 2019). In 2019, the UK became one of the first advanced economies to pass legislation to target net zero greenhouse gas emissions by 2050 (Department for Business, Energy & Industrial Strategy, 2019). Prior to the disruption caused by the COVID-19 pandemic, the UK was projected to miss its previously set 2030s fourth and fifth carbon budgets even though it overachieved the targets set for 2020 (Committee on Climate Change, 2019b).

Direct and indirect GHG emissions from residential building sector accounted for 16% of the UK total emissions in 2019 (Committee on Climate Change, 2020a). A recent report by the Committee on Climate Change points out that achieving the net zero emissions goal is all but impossible without decarbonizing England's existing residential stock of almost 25m homes (Committee on Climate Change, 2019a). Between 1990 and 2004 the residential stock saw next to no reduction in direct carbon dioxide emissions, and between 2004 and 2019 it saw about a 25% reduction (Committee on Climate Change, 2019a). In England, more than half the stock has an energy efficiency band of D (rated from A-G, with A being the top rating) or worse with a large portion of the housing stock built before 1980 (Henretty, 2020). Decarbonisation of the English stock can be achieved by a range of measures including switching to decarbonised electricity, repurposing the existing gas network for green hydrogen, district heating with renewables, and improving the efficiency of the building fabric, heating systems, lights and appliances. The significant size and age of the stock means that, in addition to its importance to the decarbonisation targets, building retrofit can also provide a significant opportunity to boost economic activity by increasing employment, and wages (Nieto et al., 2020). Although there are many options for achieving net zero, practically delivering this transition will pose a number of challenges, including, identifying the stock most in need of

intervention, the deployment of the right mix of intervention measures that are practical, and most cost effective, for both home owners and the whole energy system.

The recently released UK Energy Performance Certificate (EPC) action plan aims to make energy performance data more accurate, reliable, and accessible by 2021 in order to better plan for the net zero transition (Department for Business, Energy & Industrial Strategy & Ministry of Housing, Communities & Local Government, 2020). Furthermore, the UK Government's Green Industrial Revolution aims to reduce carbon emissions by 71MtCO_2e between 2023-2032 by installing 600 thousand heat pumps annually by 2028, improving the energy efficiency of 2.8 million houses by 2030, and ensuring 'zero carbon ready' new build construction (Department for Business, Energy & Industrial Strategy, 2020c). Concurrently, the government has made a pledge to invest in offshore wind capacity such that every home in the country can be wind powered by 2030 (Prime Minister's Office, 2020).

Many aspects of residential building retrofit have already been explored within the literature from financing and governance requirements, to the development of various scenario modelling tools to support decision making in the area. Ahlrichs and colleagues (2020) have studied retrofit investment and financial barriers highlighting a need in increased emission taxes to influence retrofit-related investment decisions. Meanwhile, a recent study of British landlords' behavior relating to retrofit activity, points at diversifying information channels targeted at landlords. This is seen as a key factor in ensuring policy and financial incentives are adopted despite differing financial means and attitudes among landlords (Miu & Hawkes, 2020). Decision support studies have also looked at susceptibility of various geodemographic groups of home-owners to invest in retrofit measures (Trotta, 2018) and the potential unintended consequences of thermal retrofit resulting in a consumption rebound (Galassi & Madlener, 2018). While, particularly in a UK context, studies have focused on identifying retrofit actions required to meet decarbonisation targets (Delmastro et al., 2016; Gillich et al., 2019; Broad et al., 2020) or the potential role for various public and private stakeholders to

enable these (Killip & Owen, 2020; Wade et al., 2020), consideration of the whole-life carbon associated with these retrofit interventions and hence the feasibility of their deployment, whilst remaining within the emissions budget has not received similar attention.

The aim of this paper is thus to investigate the whole-life carbon implications of residential retrofit. Starting from a series of retrofit measures to be deployed across the housing stock in England, we first estimate a carbon budget for the residential sector leading to the year 2050. We then model the operational emissions along with the embodied emissions associated with each measure. The combined emissions over the period 2021-2050 is then compared against the developed budget to determine feasibility of retrofit scenarios in meeting the budget or the additional renewables required to offset budget overshoots. The results suggest that there in fact exist specific mixes of retrofit interventions that can be deployed within the national carbon budget by 2050. The scale required for the deployment of these measures, however, is likely to pose enormous practical challenges, with these intensifying with lack of action as we get closer to 2050.

2 Methodology

2.1 Retrofit scenarios

We consider four main types of retrofit measures in this study: wall insulation, glazing replacement, PV modules and heating system upgrade. They are applied individually and combined. In the combined retrofit measures, only double glazing is considered due to its higher financial effectiveness in retrofitting UK residential dwellings when compared with triple glazing (Salem et al., 2020). The retrofit scenarios considered in this study are detailed in Table 1. The various measures under consideration are applied to the building stock in such a way that the definition of the measure and the homes to which it could be applied are explicitly programmed. For example, cavity wall insulation is defined as providing a U-value of 0.1 W/(m²·K) after installation and is only applied to homes with cavity walls currently without cavity insulation. The National Household Model then searches the stock for suitable homes

and applies the measure as defined. Table 2 provides information on the aggregate scope of each scenario when applied to all eligible stock across the nearly 25m units available to retrofit.

Table 1. Retrofit scenarios and the description of the interventions modeled for the stock.

Retrofit Scenario	**Description**
BL	Baseline (BL), no retrofit, the stock stays as it is currently without upgrade.
PV	Add photovoltaic (PV) modules on all buildings with available roof space for PV. The PV modules are assumed to be installed on 30% of all available building roof space, with an efficiency of 20% for electricity generation. All PV has an encumbered view of the sky according to the orientation of the building
CI	Cavity insulation (CI) applied to all buildings with cavity walls where no previous cavity wall insulation is installed to achieve a post retrofit U-value of 0.1 W/(m²·K) .
EI	External insulation (EI) applied to all buildings with solid walls and no previous external wall insulation is installed, to achieve a post retrofit U-value of 0.1 W/(m²·K) .
II	Internal insulation (II) applied to all buildings of any wall type where no previous internal wall insulation was installed, to achieve a post retrofit U-value of 0.1 W/(m²·K)
DG	Replace windows in homes where the glazing U value is greater than 3 W/(m²·K) to double glazing (DG) with U-value of 1.6 W/(m²·K).
TG	Replace windows in homes where the glazing U value is greater than 2 W/(m²·K) to triple glazing (TG) with U-value of 0.8 W/(m²·K) .
GSHP	Switch to ground source heat pump (GSHP) with seasonal COP of 3 if outdoor space is available and the current heating system efficiency is less than 100% or uses community heating.
ASHP	Switch to air source heat pump (ASHP) with seasonal COP of 2.5 if the current heating system efficiency is less than 100% or uses community heating.
HBHP	Switch where the current heating system efficiency is less than 100% and the house is connected to the gas network to hybrid heat pump (HBHP) with 50:50 split of heating supply.
AWI	All Walls Insulated (AWI). Apply CI, EI, and II sequentially to all the stock with the same rules as above. No individual home receives more than a single insulation measure to achieve a post retrofit U-value of 0.1 W/(m²·K) for the treated walls. Many homes which were not classified as simple CI or EI suitable homes and have mixed wall and layout types will result in non-uniform application of insulation measures resulting in a higher than 0.1 W/(m²·K) for the dwelling.
AWI+DG+ASHP	Combining the all wall insulation, glazing and heating system switch to air source heat pump or hybrid heat pump as per the listed combinations, the measures are applied to the stock as per their individual rules.
AWI+ASHP	
AWI+DG	
DG+ASHP	
AWI+HBHP	
DG+HBHP	
AWI+DG+HBHP	

In practice heat pumps vary significantly in operational performance (seasonal COP of 1.5 to 4.5 for ASHPs), due to installation quality, product performance and maintenance, with this issue well documented (UCL Energy Institute & Department for Business, Energy & Industrial Strategy, 2017). This variation in COPs is much greater than the variance in boiler efficiency, resulting in a greater variance in energy demand. At-the-meter energy consumption of the stock as a whole is modelled before and after intervention measures in order to assess the impact of installing each intervention, see online supplementary information for detailed model output. The energy demand by fuel from all energy vectors was also collected to gain an accurate picture of the operational carbon resulting from the intervention measure. The main driver of demand within SAP energy calculations (Stone et al) was identified as the heating system efficiency. Therefore, we capture the sensitivity of the analysis to variation in seasonal COP across a range from 1.5 to 3.5.

Table 2. Retrofit measures deployment scope across the eligible English housing stock.

Retrofit Scenario	**PV module area (m²)**	**Wall insulation total applied area (m²)**			**Glazing replacement total applied area (m²)**		**Annual generated thermal energy for domestic space heating from the retrofitted heating system (kWh_t)**		
		Cavity	**External**	**Internal**	**Triple glazing**	**Double glazing**	**GSHP**	**ASHP**	**Boiler**
PV	3.89E+08								
CI		9.52E+08							
EI			6.36E+08						
II				1.65E+09					
TG					4.06E+08				
DG						1.12E+08			
GSHP							1.90E+11		
ASHP								2.07E+11	
HBHP								9.25E+10	9.25E+10
AWI		9.52E+08	6.36E+08	5.90E+07					
AWI+DG+ASHP		9.52E+08	6.36E+08	5.90E+07		1.12E+08		1.37E+11	
AWI+ASHP		9.52E+08	6.36E+08	5.90E+07				1.42E+11	
AWI+DG		9.52E+08	6.36E+08	5.90E+07		1.12E+08			
DG+ASHP						1.12E+08		2.03E+11	
AWI+HBHP		9.52E+08	6.36E+08	5.90E+07				5.46E+09	5.46E+09
DG+HBHP						1.12E+08		9.06E+10	9.06E+10
AWI+DG+HBHP		9.52E+08	6.36E+08	5.90E+07		1.12E+08		4.39E+09	4.39E+09

2.2 Long-term targets and emissions budget

Based on a UK carbon budget developed by the Committee on Climate Change (CCC) (Committee on Climate Change, 2020b) and one by the Tyndall Centre (Tyndall Centre, 2020), we develop two budgets specifically for the English residential building stock, b_p, for the period p following

$$b_p = B_p * F_{be} * F_d * F_e \qquad (1)$$

where B_p is the CCC or Tyndall-developed carbon budget for UK for the period p; F_{be} is the fraction of carbon emissions from UK built environment to UK total carbon emissions, which was around 40% in 2019, including embodied carbon emissions from new construction, operational carbon emissions from existing built assets and direct emissions from transportation (UK Green Building Council, 2020); F_d is the fraction of UK residential carbon emissions (including both residential embodied and operational carbon emissions) to the UK built environment carbon emissions (considering embodied and operational carbon emissions of three sectors, namely residential buildings, non- residential buildings and infrastructure), which was 61% in 2010 (Green Construction Board, 2013); F_e is the fraction of England's residential carbon emissions to the UK residential carbon emissions which is calculated based on the prediction of Tyndall carbon budget between that for England and the UK (as shown in 3).

The Tyndall carbon budget is more stringent than the CCC carbon budget, as the Tyndall interpretation of the Paris agreement is that developed nations should be apportioned a smaller proportion of the global carbon budget, with developing nations given a larger share to facilitate their development. In addition, global elements of the budget are also allocated to areas such as land use change, see Anderson et al (2020, and Kuriakose et al. (2021) for further details. The CCC carbon budgets cover 5-year periods and are set every five years, with the sixth CCC carbon budget (2033-2037) recently published in December 2020. The Tyndall carbon budgets have been published and also cover five-year periods from 2018 to 2047 with additional budgets covering 2048 to 2100. The total budget from 2021 to 2050 is then disaggregated assuming a linear proportionality across the years during the periods 2018-2022 and 2048-2100 according to equation 2.

$$b_{2021-2050} = \frac{2}{5} b_{2018-2022} + b_{2023-2027} + b_{2028-2032} + b_{2033-2037} + b_{2038-2042} + b_{2043-2047} + \frac{3}{53} b_{2048-2100} \quad (2)$$

Table 3 shows the extracted carbon budgets from CCC and Tyndall for the UK and the scaled budget for the English residential stock. It is worth mentioning that the current indicative path to the UK's net zero target shows the annual GHG emissions will not reach this target before 2050 (Committee on Climate Change, 2020c), and the CCC carbon budget for 2033 to 2037 has only just been published (Committee on Climate Change, 2020d). Therefore, our estimated CCC England carbon budget for the period 2021-2050 is likely to be an underestimation. Also, the Tyndall carbon budgets are for carbon dioxide emissions only instead of in carbon equivalents. With this study focusing on how to successfully mitigate England's residential stock carbon emissions, whilst keeping to the 2050 carbon budget, we only consider the total carbon budget between 2021-2050 as an aggregated benchmark instead of trying to meet intermediate periodical 5-year targets. The study takes a consumption-based approach (Bhattacharya et al., 2020) to carbon accounting, including the embodied emissions from all materials whether they are produced in the UK or internationally.

Table 3. Carbon budgets for the UK and England residential sector – note that only CCC values are in carbon equivalent.

Period	**CCC UK carbon budget ($MtCO_2e$)**	**Tyndall UK carbon budget ($MtCO_2$)**	**Tyndall England carbon budget ($MtCO_2$)**	F_e **the fraction of England's residential carbon emissions to the UK residential carbon emissions**	**England residential carbon budget**	
					CCC ($MtCO_2e$)	**Tyndall ($MtCO_2$)**
2018-2022	2544	1456.9	1175.6	80.7%	500.9	286.8
2023-2027	1950	741.0	599.6	80.9%	385.0	146.3
2028-2032	1725	366.3	297.3	81.2%	341.6	72.5
2033-2037	965	181.3	147.7	81.5%	191.8	36.0
2038-2042	/	89.9	73.5	81.8%	/	17.9
2043-2047	/	44.9	36.9	82.2%	/	9.0
2048-2100	/	44.4	36.8	82.9%	/	9.0
2021-2050	**4692.6**	**2008.7**	**1627.3**		**1118.8**	**397.1**

2.3 Modeling operational carbon

2.3.1 National Household Model

The National Household Model (NHM) is an open-source modelling tool developed for the UK Department of Business Energy and Industrial Strategy (2017). The model (v 7.2.2 used in this study) brings together the standard monthly building energy assessment calculation methods with the representative UK residential building stock. The energy assessment models available are the Standard Assessment Procedure (SAP) and Building Research Establishment Domestic Energy Model (BREDEM), which are closely related in terms of being monthly energy balance methods and SAP having been derived from BREDEM to fulfil the role of the National Calculation Model for the UK to conform to the EU Energy Performance of Buildings Directive (BRE, 2013, 2014). For the purposes of calculating operational carbon here, we use the BREDEM 2012 energy calculator.

The NHM requires building stock data and has been designed to take English Housing Survey (EHS) data as a primary building stock. For the purposes of this analysis, we use data from the EHS 2012 (Ministry of Housing, Communities & Local Government, 2014). Although the NHM has the capability to perform scenario modelling over time, with programmable interventions and changes to the building stock, the operational carbon calculations in this analysis are done over one calendar year with all interventions implemented on the first day of the model run. This is because we are primarily interested in investigating the combined embodied and operational carbon implications of retrofit measures and a deploy-all-first approach represents the most favorable approach in minimizing accumulating operational emissions in the absence of practical constraints.

The annual energy consumption of the whole English housing stock after undertaking different retrofit measures is presented in Table 4. The context of energy savings from the different passive fabric measures is such that the comparison across measures is not immediately clear. Wall insulation measures are applied to the appropriate wall types, with the target U value of 0.1 W/(m²·K), which means that the scale of energy savings depends solely on the number of houses treated and

the original U value. For glazing measures, the comparison is complicated by the variable ‘before’ and ‘after’ U values of the windows as well as a variable effect on the solar gains in each building. Unlike with wall insulation measures, the ‘after’ U value varied between DG and TG is significantly different (1.6 vs 0.8 W/(m²·K)) as well as the different ‘before’ U values due to DG measure only being applied to single glazed properties and TG being applied to both single- and double-glazed homes. When applied across the non-homogenous stock, these differences make direct comparisons across similar measures complex, and conclusions on the implications for individual homes are not possible from the building stock aggregate results. Detailed output from the model for each scenario is available within the supplementary material.

Table 4. The energy consumption of the English house stock under different retrofit scenarios

Retrofit Scenario	Annual estimated energy consumption (TWh)							Annual estimated PV generated electricity fed back to grid (TWh)
	Grid electricity	Community[a]	Natural gas	Oil	LPG	Coal	Biomass	
BL	99.27	2.89	338.07	23.82	6.79	2.51	1.76	-
PV	73.77	2.89	338.07	23.82	6.79	2.51	1.76	27.86
CI	96.91	2.60	310.83	21.78	6.20	2.36	1.72	-
EI	96.32	2.67	296.89	19.69	5.65	2.00	1.31	-
II	93.81	2.36	267.21	17.55	5.02	1.81	1.27	-
TG	98.53	2.64	316.88	22.48	6.32	2.37	1.65	-
DG	99.16	2.83	334.45	23.50	6.69	2.46	1.72	-
GSHP	173.70	2.89	60.52	0.17	4.04	0.13	0.03	-
ASHP	197.42	2.89	32.21	0.01	4.01	0.12	0.03	-
HBHP	132.01	2.89	196.04	22.98	6.39	2.07	1.25	-
AWI	93.81	2.36	267.21	17.55	5.02	1.81	1.27	-
AWI+DG+ASHP	168.70	2.28	24.89	0.00	2.65	0.09	0.02	-
AWI+ASHP	170.20	2.36	25.35	0.00	2.75	0.09	0.02	-
AWI+DG	93.64	2.28	262.75	17.14	4.89	1.75	1.20	-
DG+ASHP	196.25	2.83	31.83	0.01	3.93	0.12	0.03	-
AWI+HBHP	95.24	2.36	220.61	16.88	4.65	1.46	0.89	-
DG+HBHP	131.43	2.83	194.23	22.67	6.28	2.03	1.22	-
AWI+DG+HBHP	94.96	2.28	217.00	16.49	4.52	1.41	0.83	-

[a] *Community as an energy consumption is technology agnostic and accounted for in same manner as the other vectors, in terms of metered energy at the building boundary. Therefore, it is classified as a vector in its own right rather than being accounted for in the Grid electricity or natural gas values, which are exclusive of Community heat demand*

2.3.2 Grid emissions and decarbonisation scenarios

To arrive at emissions related to residential consumption we need to assume a carbon intensity for the national grid in the period 2021-2050. From 2008 to 2019, the carbon emissions from the power sector fell by 67% with the decarbonisation of the power sector dominating the UK carbon emissions fall for the last decade (Committee on Climate Change, 2020c). However, the power sector needs to be further decarbonised to achieve the net zero carbon emissions target. In this study, the effect of grid decarbonisation on the carbon emissions factor of electricity is explored by initially considering six future grid decarbonisation scenarios based on the analysis of Committee on Climate Change and that of the national grid:

S1: In 2019, the electricity carbon emissions for the UK were 200 gCO_2/kWh (Department for Business, Energy and Industrial Strategy, 2020). According to the analysis and targets set by the Committee on Climate Change (2019b), the grid carbon emissions are assumed as 50gCO_2/kWh in 2030 and 10gCO_2/kWh in 2050. Interpolation is used to calculate the grid carbon emissions of other years in-between, with a constant decarbonisation pace assumed between 2019-2030 and 2030-2050.

S2 to S5: The national grid (2020) has developed a set of four future energy scenarios with different CO_2 intensities of electricity generation, namely consumer transformation (S2), system transformation (S3), leading the way (S4) and steady progression (S5). The three scenarios, S2-S4, all rely on bioenergy with carbon capture and storage (CCS) and achieve negative emissions prior to 2050.

S6: In the Committee on Climate Change's sixth carbon budget (Committee on Climate Change, 2020d), a balanced net zero pathway for electricity generation was developed, which very largely decarbonises electricity generation by 2030, and decarbonises it completely by 2035, action thereafter focuses on meeting rising demand with low-carbon generation.

These grid carbon emissions scenarios only account for direct CO_2 emissions. In this paper, we make adjustments to include all GHG emission in the grid emissions. By assuming the ratio between grid emitted GHG and CO_2 remains constant at its 2017 value of 1.008 (Department for Business, Energy and Industrial Strategy, 2019), the emission factor of electricity accounting for all GHG is calculated following:

$$OC_{e,y}=G_{CO_2,y} * 1.008 \quad (3)$$

$OC_{e,y}$ is the carbon emission factor of electricity in year y in $kgCO_2e/kWh$; $G_{CO_2,y}$ is the grid carbon emissions considering only CO_2 in year y measured in gCO_2/KWh. The grid carbon emissions of all scenarios from 2019 to 2050 are presented in Figure 1. The grid carbon emissions in 2019 are lower for national grid scenarios (S2 to S5) as the analysis of the national grid ignores network and operability constraints on the transmission or lower voltage networks (National Grid, 2020).

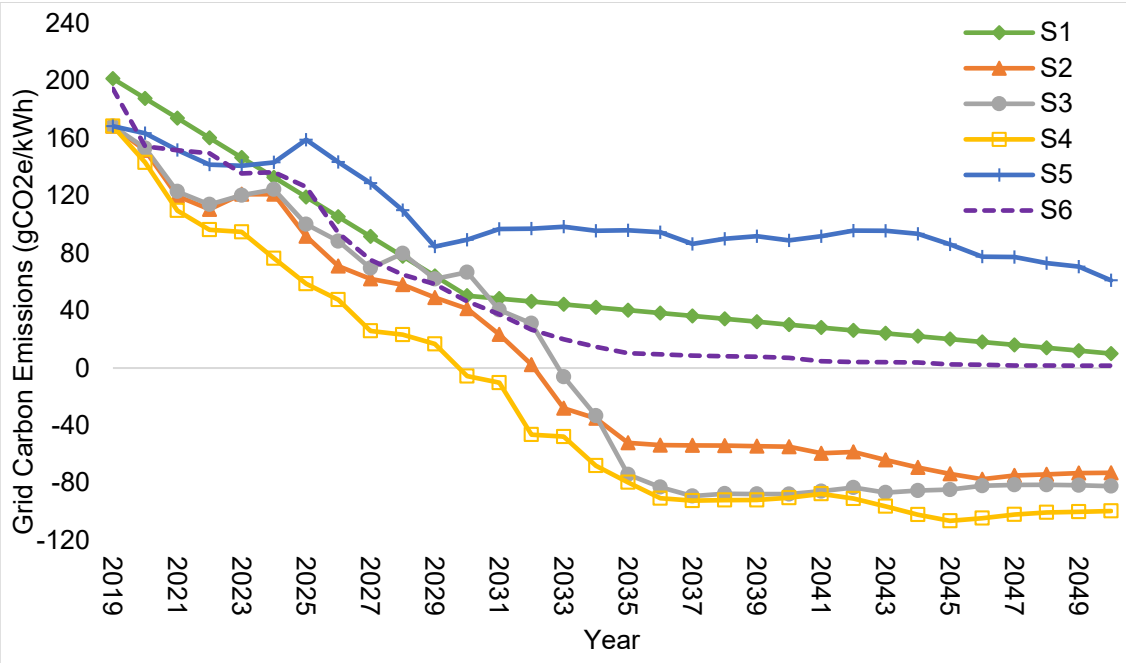

Figure 1. Future grid carbon emissions for all scenarios

Since a successful and scalable deployment of CCS before 2050 could ensure a net zero transition almost independently from other available interventions, we will only use scenarios S1, S5 and S6

in our grid emissions results moving forward. Results from the other three future grid carbon emissions scenarios can be accessed in the supplementary material. It is also worth considering, that given current progress on CCS scalability and deployment, that CCS mass deployment is an unlikely scenario before 2050 (Bui et al., 2018).

2.3.3 Operational carbon

Finally, the total annual operational carbon emissions for 2021-2050, O, is then calculated based on the operational energy consumption after:

$$O = \sum_{2021}^{2050} O_y = \sum_{2021}^{2050} [(M_e - P) \times OC_{e,y} + M_g \times OC_g + M_o \times OC_o + M_l \times OC_l + M_c \times OC_c + M_m \times OC_m + M_b \times OC_b] \quad (4)$$

where M is the annual consumption of a specific fuel in kWh; P is the annual photovoltaic electricity generation fed back to grid in kWh; OC is the carbon emission factor of a specific fuel in gCO_2e/kWh, with the subscript e, g, o, l, c, m, and b standing for electricity, natural gas, oil, LPG, coal, community, and biomass, respectively. The values for OC are taken as $OC_g = 184$, $OC_o = 247$, $OC_l = 214$, $Oc = 345$, $OC_m = 50$, $OC_b = 16$ (Department for Business, Energy and Industrial Strategy, 2019; Department for Energy and Climate Change, 2015). Although in the future, more sustainable fuels, e.g. biogas, hydrogen might be able to replace some portion of the carbon intensive fossil fuels currently consumed within residential stock, the approach taken in this study, using the NHM model is based on SAP methodology, which utilises present-day energy vectors and does not take direct account of innovative fuels such as biogas and hydrogen. As such, the implications of these fuels is excluded from the analysis, both in terms of residential demand, as well as, implications such as additional grid generation for fuel production. Therefore, constant carbon emission factors are assumed for natural gas, oil, LPG, coal, community and biomass between 2021 and 2050.

2.4 Embodied carbon calculation

Embodied carbon measures greenhouse gas emissions during the non-operational phase and as such includes emissions released through extraction, manufacture, transportation, assembly,

maintenance, replacement, deconstruction, disposal, and end-of-life aspects of the materials and systems that make up a building (UK Green Building Council, 2015). The scope of embodied emissions in this study covers cradle-to-gate, i.e., A1-A3 embodied carbon (RICS, 2017). This essentially covers the supply-chain up to the point that construction products are ready for use at the factory gate.

2.4.1 Retrofit embodied carbon

Insulation: We consider three types of wall insulation in this study. These are solid wall external insulation, internal insulation and cavity wall insulation. Thirteen different insulation materials have been considered in this study to estimate mean embodied emissions across available options. (See supplementary material for the characteristics of the individual insulation materials used.) The thickness of insulation material to meet the required U-value specified in Table 1 is calculated using

$$\delta = \lambda \times (1/U_b - 1/U_a) \qquad (5)$$

where U_b and U_a are the thermal transmittance of the wall before and after the retrofit in W/(m²·K); δ is the thickness of chosen insulation material in meters; and λ is the thermal conductivity of chosen insulation material in W/(m·K). The embodied carbon of wall insulation, E_w, can then be calculated following

$$E_w = \delta \times A_w \times D \times EC_w \qquad (6)$$

where A_w is the total area of retrofitted wall in m^{2,} see Table 2; D is the density of the chosen insulation material in kg/m^3; and EC_w is the cradle-to-gate embodied carbon intensities of the insulation measured in kgCO_2e/kg.

Glazing: We consider double and triple glazing as the two types of market available retrofit measures for openings. The embodied carbon of window replacement, E_d, is calculated following

$$E_d = EC_d \times A_d \qquad (7)$$

where A_d is the retrofitted window area in m^2 and EC_d is the cradle-to-gate embodied carbon intensities of the replaced window taken as 32.5 $kgCO_2e/m^2$ for double glazing and 52.4 $kgCO_2e/m^2$ for triple glazing, respectively (Jones & Hammond, 2019).

Heat pump and boiler: In the UK, governmental renewable heat schemes like Domestic Renewable Heat Incentive (RHI) have historically promoted the deployment of heat pumps (GOV.UK, 2015). Air source, ground source, and hybrid heat pumps, i.e., combined air source heat pump and condensing gas boiler, are considered to replace existing boilers in eligible stocks. The embodied carbon of heating system replacement, E_h, is calculated through

$$E_h = EC_h \times T_h + EC_b \times T_b \qquad (8)$$

where EC_h is the cradle-to-gate embodied carbon intensities of heat pump in $kgCO_2e/kWh_t$; T_h is the thermal energy generated by heat pumps for residential space heating and hot water in kWh_t; EC_b is the cradle-to-gate embodied carbon intensities of condensing gas boilers in $kgCO_2e/kWh_t$; and T_b is the thermal energy generated by boilers for residential space heating and hot water measured in kWh_t. Based on a study by Greening and Azapagic (2012), assuming that 95% of Global Warming Potential (GWP) impact is from the operational energy consumption, the life cycle embodied carbon intensities of air source heat pumps, ground source heat pumps and condensing gas boilers are taken as 0.014$kgCO_2e/kWh_t$, 0.009$kgCO_2e/kWh_t$, and 0.015$kgCO_2e/kWh_t$, respectively. Including extraction and processing of fuels and raw materials; system manufacture; installation; maintenance; decommissioning and transportation for heat pumps, as well as extraction and processing of fuels and raw materials; boiler manufacture; decommissioning and transportation for boiler. Given that these values include emissions beyond cradle-to-gate, such as installation and disassembly, our values for these components are likely to lead to a slight overestimation of their cradle-to-gate embodied carbon.

Photovoltaics: We calculate the embodied carbon of the PV modules, E_p, following

$$E_p = EC_p \times A_p \qquad (9)$$

where A_p is the area of PV modules installed in m^2 and EC_p is the cradle-to-gate embodied carbon intensities of PV modules taken as 242 kgCO_2/m^2 for monocrystalline PV modules, 208 kgCO_2/m^2 for polycrystalline PV modules, and 67 kgCO_2/m^2 for thin-film PV modules (Jones & Hammond, 2019). The embodied carbon intensities of PV modules only accounts for CO_2 instead of all GHG meaning the overall embodied carbon of PV modules are generally underestimated.

3 Results and discussion

3.1 Whole-life carbon emissions from retrofit

The embodied carbon emissions of applying the aforementioned retrofit measures for all suitable residential buildings in the English housing stock is presented in Figure 2. The error bars for embodied emissions account for the variation in material for insulation and PV installation as more than one material and one type of PV module are considered. It is clear that retrofit scenarios which include installation of insulation, especially those of external and internal type, can have a wide range of embodied emissions highlighting the importance of material selection. For example, when using wood fibre as insulation material, real-terms emission saving can be achieved as a result of the overall sequestration of carbon within the product.

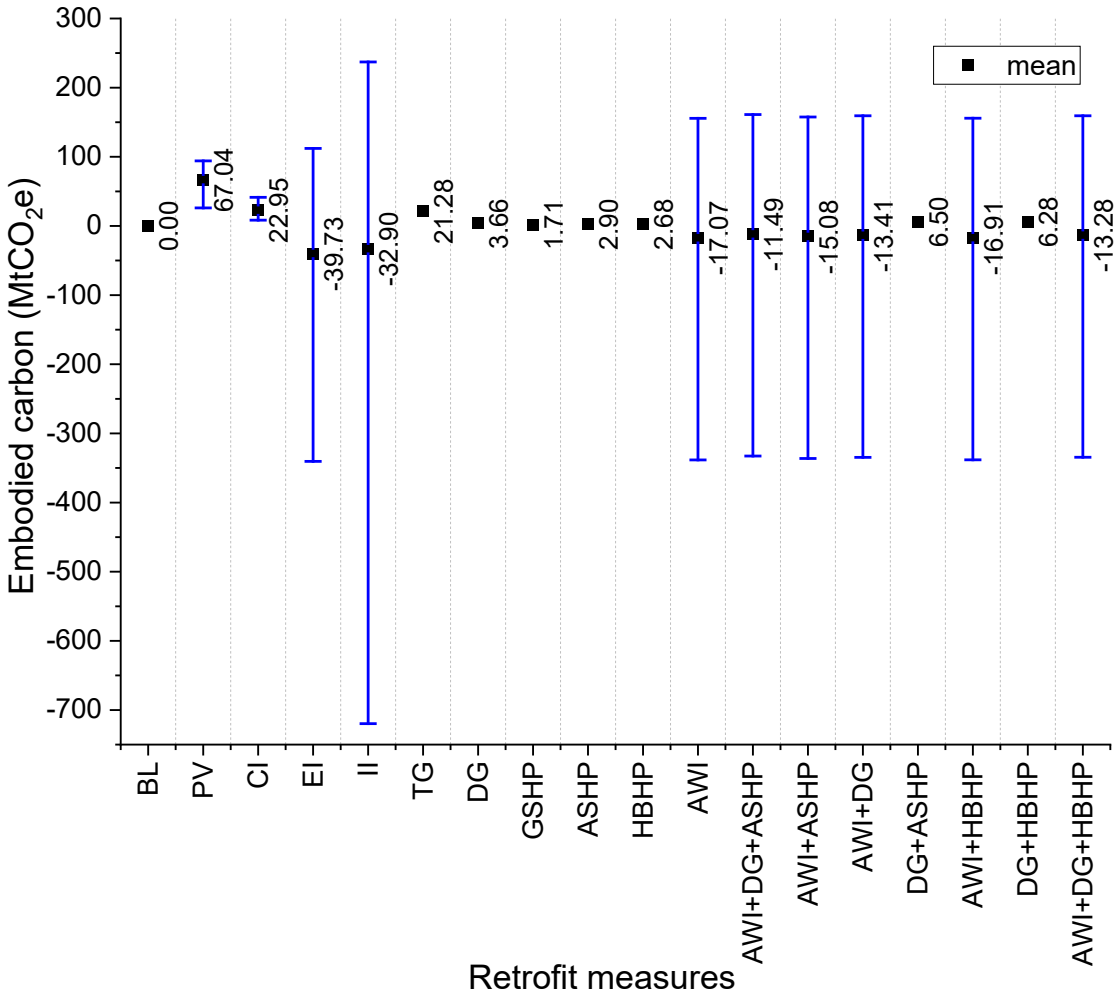

Figure 2. Embodied carbon emissions of different retrofit scenarios

Figure 3, shows the corresponding operational carbon emissions over a 30 year period up to 2050 and how these operational emissions relate to the Tyndal and CCC carbon budgets assuming that all the measures were undertaken at the beginning of 2021 and that the grid decarbonises at three different rates/scenarios. The one single measure applied on its own that has the biggest impact on operational carbon emissions is the ASHP. However, this has to be combined with insulation measures I,e, AWI+DG+ASHP and AWI +ASHP under S6, to meet the strictest Tyndall carbon budget. When compare with the baseline of undertaking no retrofit at all, the operational carbon emissions savings of all retrofit scenarios considered ranges from 1% for DG to 81% for AWI+DG+ASHP under S1 grid decarbonisation scenario, 1% to 72% under scenario S5, and 1% to

84% under scenario S6. What is crucial to note is that, without a mass deployment of air or ground source heat pumps, the 30-year operational carbon emissions from any of the retrofit scenarios would exceed both CCC and Tyndall carbon budgets.

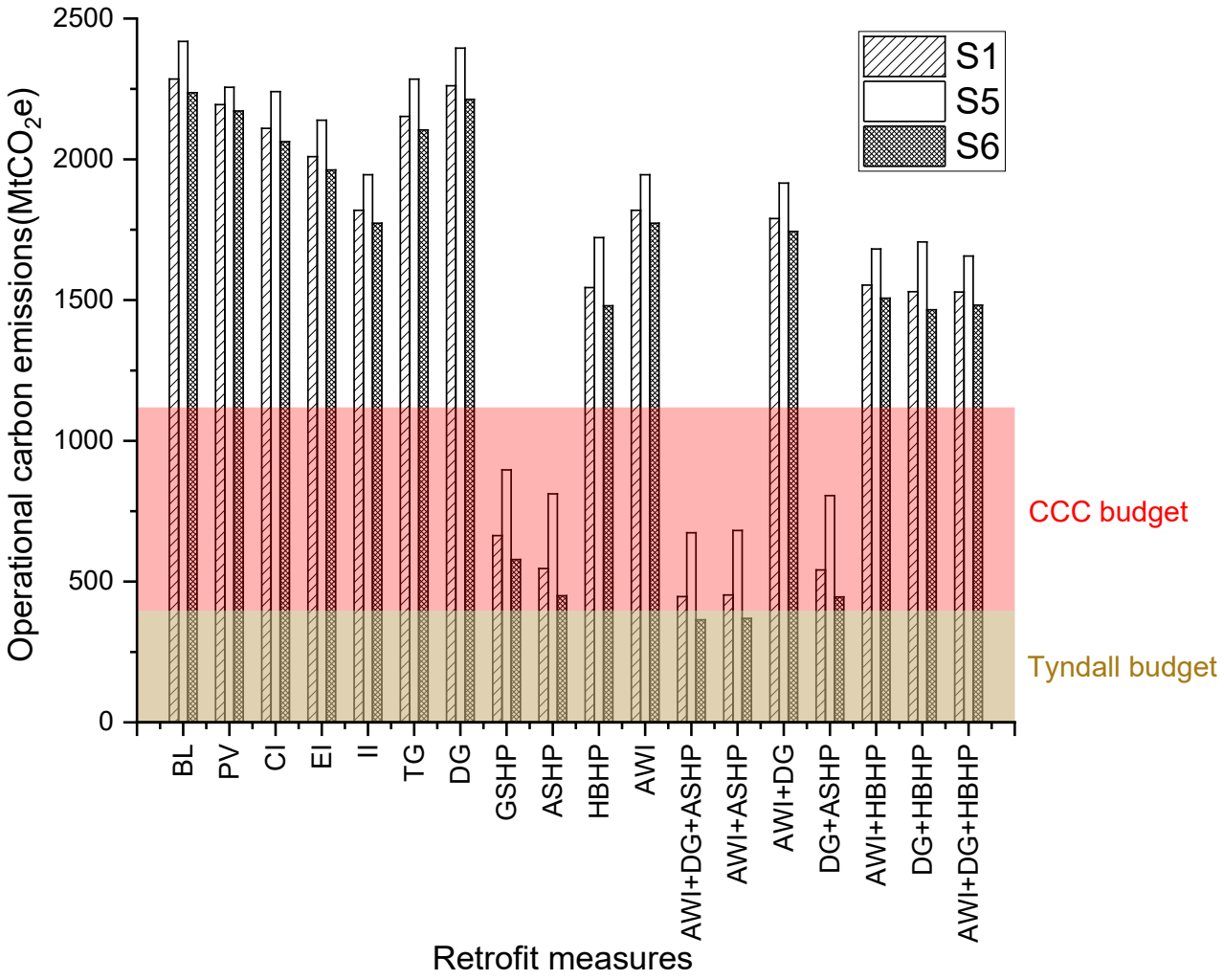

Figure 3. 30 years (2021-2050) operational carbon emissions of different retrofit scenarios – highlighted regions show the CCC and Tyndall budgets between 2021-2050.

Compared with the 30 years of post-retrofit operational carbon emissions, however, the mean embodied carbon emissions from retrofit measures are almost negligible. Under S1, S5 and S6, the carbon payback of applying various retrofit measures is presented in Table 5, apart from PV, all scenarios achieved carbon reduction through retrofit within 5 years, while many of them emit less

carbon emissions compared to the baseline from the first year. The carbon payback for PV modules is over 7 years, and it is not able to payback at the end of our study period (30 years from 2021 to 2050) under S6 grid decarbonisation scenario. It's of interest to note that both double glazing and triple glazing are able to achieve carbon payback within 5 years when applied alone. Although the cradle-to-gate embodied carbon intensities for triple glazing is 1.6 times of that of double glazing, the triple glazing achieved bigger energy savings as more properties can accommodate this upgrade.

Table 5.The carbon payback time [unit: year(s)]

Retrofit scenario	**S1**	**S5**	**S6**
PV	13	10	>30
CI	4	4	4
EI	1	1	1
II	1	1	1
TG	5	5	5
DG	5	5	5
GSHP	1	1	1
ASHP	1	1	1
HBHP	1	1	1
AWI	1	1	1
AWI+DG+ASHP	1	1	1
AWI+ASHP	1	1	1
AWI+DG	1	1	1
DG+ASHP	1	1	1
AWI+HBHP	1	1	1
DG+HBHP	1	1	1
AWI+DG+HBHP	1	1	1

As mentioned, the selection of low environmental impact materials and systems in retrofit can significantly reduce embodied carbon emissions, as is the case with the external and internal wall insulation. Considering total whole-life carbon emissions, as shown in Figure 4, careful selection of materials and systems used in retrofit provide a greater chance to meet stricter carbon budgets. However, there must also be a focus on ensuring the materials to perform well in situ, i.e. to reduce the performance gap (Gupta & Kotopouleas, 2018).

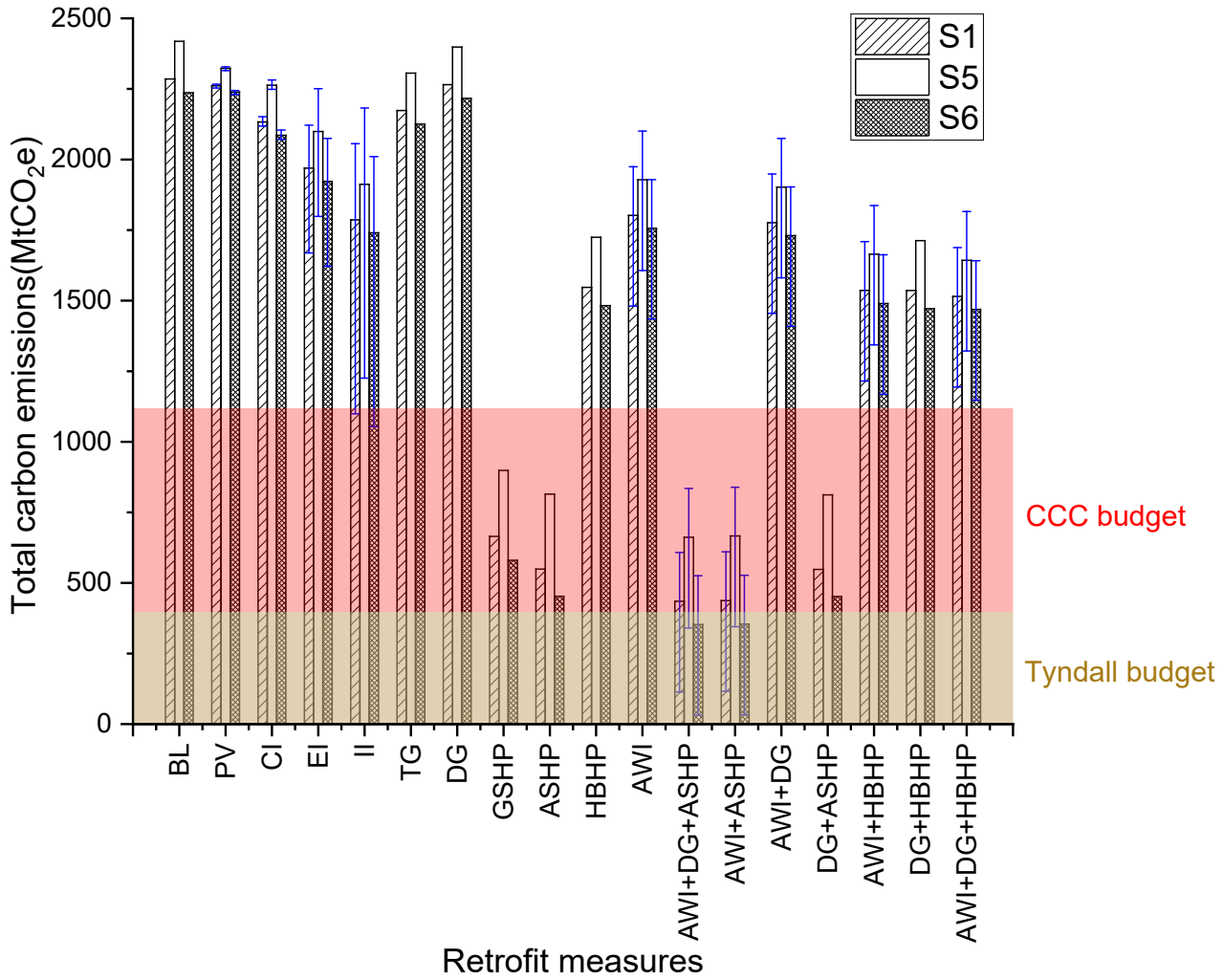

Figure 4. 30 years (2021-2050) total embodied and operational carbon emissions when considering different retrofit measures and grid decarbonisation scenarios for English residential stock – note that heat pumps are the only way that the budgets can be met with the Tyndall budget requiring careful selection of insulation material across the entirety of the stock.

3.2 England's house stock carbon emissions in 2050

The English housing stock will play a key role on whether net zero emissions can be achieved in the UK by 2050. The operational carbon emissions per annum in 2050 for different grid decarbonisation scenarios and retrofit combinations are presented in Figure 5, by using electricity carbon emission factors in 2050. Based on the stock operational energy prediction results from NHM, the English

housing stock is not likely to achieve net zero carbon emissions by 2050 through building retrofit measures alone even when considering non-transformational grid decarbonisation. Achieving net zero in 2050 as such requires either the successful deployment of CCS technologies or a massive increase in renewables capacity, likely to be wind, by 2050.

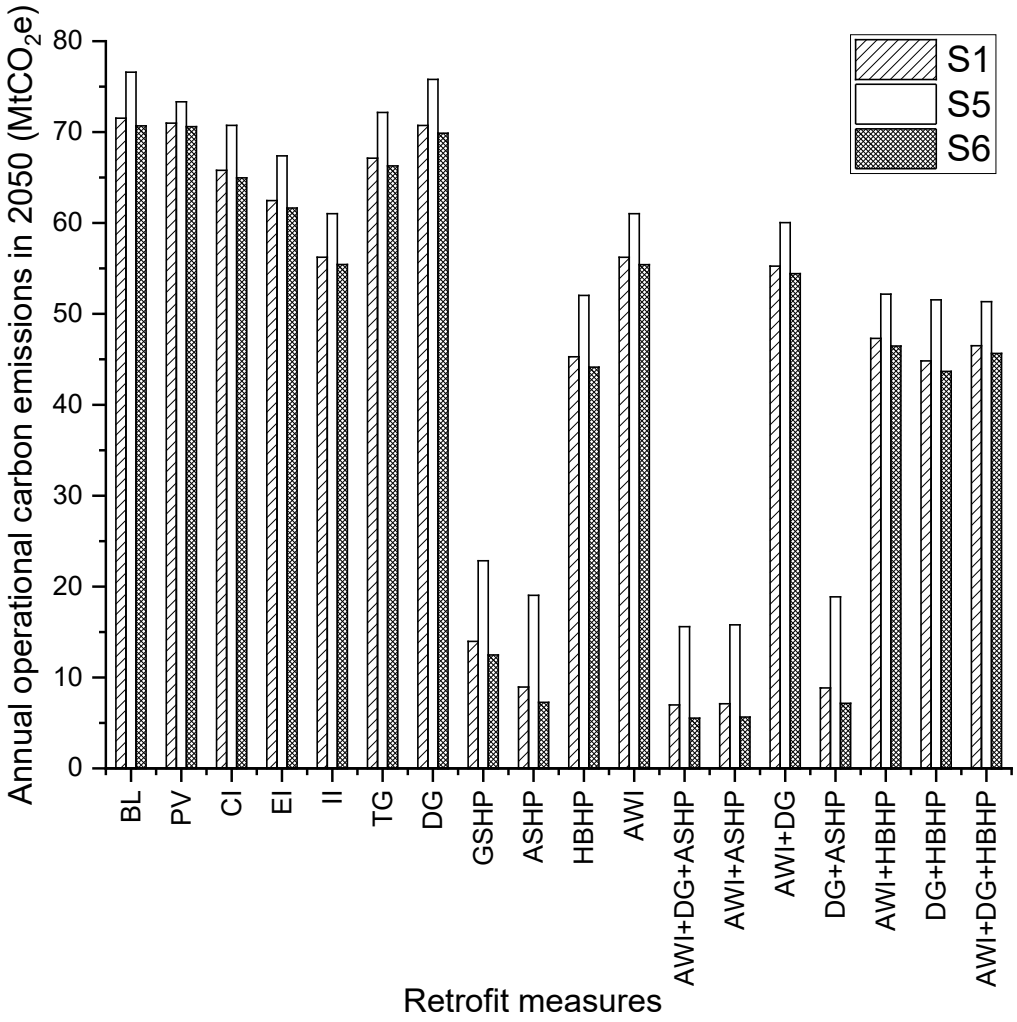

Figure 5. Operational carbon emissions in 2050 (one year) for different grid decarbonisation and retrofit scenarios.

3.3 Additional wind capacity to meet carbon budget

UK wind capacity has continuously increased from 5.4 GW in 2010 to 24 GW in 2019. By 2019, wind made up the UK's second largest source of electricity (Department for Business, Energy & Industrial

Strategy, 2020a). Current ambitious plans to power all UK homes with wind by 2030 could theoretically provide carbon reduction benefits as well (Elgot et al., 2020). According to Smoucha and Fitzpatrick (2016), an onshore 2 MW turbine with annual electricity generation of 4520 MWh has 787 tCO_2e of embodied carbon emissions from the manufacturing stage. Under S1, S5 and S6 grid scenarios, a 2 MW turbine achieves net carbon reduction at the end of the 2nd year. Whilst 30 years operation provides for a carbon emissions offset of 6929 tCO_2e, 13007 tCO_2e and 4721 tCO_2e, respectively, when compared with the grid electricity. This we use to estimate the wind capacity required to meet Tyndall carbon budget targets.

For all the retrofit scenarios considered in this study, among those failing to meet the Tyndall carbon target, DG+ASHP has the lowest total carbon emissions. After the retrofit interventions, the 30 years (from 2021 to 2050) total carbon emissions from the English housing stock are 150.88 $MtCO_2e$, 414.79 $MtCO_2e$ and 54.99 $MtCO_2e$ over the Tyndall carbon target under S1, S5 and S6 scenarios, respectively. To help meet the Tyndall carbon target, 21,776 sets, 31,890 sets and 11,650 sets of 2MW turbines need to be installed depending on the grid scenario. The accumulated carbon emissions from the stock after DG+ASHP retrofit and accumulated carbon reductions from the wind turbines from 2021 to 2050 are presented in Figure 6.

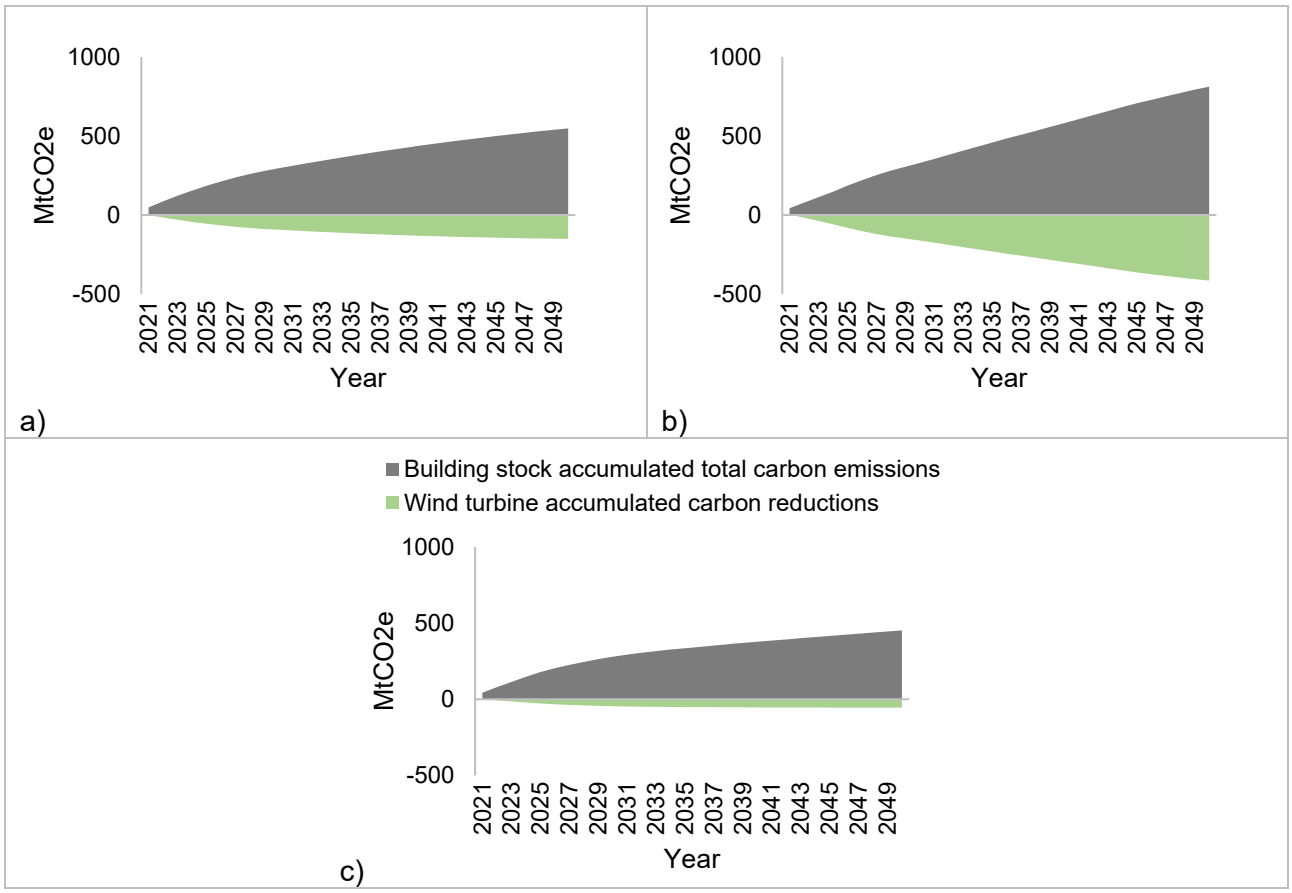

Figure 6. The carbon emissions from English house stock after DG+ASHP retrofit and carbon reduction from wind turbines between 2021 and 2050 under S1 (a), S5 (b) and S6 (c)– the accumulated carbon reduction from wind turbines are assigned with negative values to offset the accumulated carbon emissions from the stock.

With the installation of enough wind turbines, the Tyndall carbon target can be achieved. However, deploying the equivalent capacity of up to nearly 32 thousand 2 MW wind turbines would pose practical challenges to load management in the national grid, even when not considering the construction challenges in delivering such capacity. More generally, our results highlight mass deployment of heat pumps to be fundamental when trying to meet carbon targets for the residential stock in England. This on its own means that the electrification of residential demand is crucial in meeting targets. The increase of electricity demand for space heating ranges from 1.76 TWh to 82.72 TWh in 9 out of 18 retrofit scenarios utilizing heat pumps, ie, 0.5% to 23.9% of the total UK

electricity demand in 2019 (Department for Business, Energy and Industrial Strategy, 2020a) which is in itself a challenge to the grid supply balance in the UK.

3.4 Limitations

The embodied carbon calculated here covers only cradle-to-site (A1-A3) unless otherwise specified. This means our embodied carbon estimates, while low, do not include the emissions associated with the construction process stage (A4-A5), use stage (B1-B7), and end of life stage (C1-C4) of the retrofit interventions (BSI, 2011). Moreover, we have not considered circumstances under which retrofit interventions might require replacement during the 30-year study period of 2021-2050. Depending on the working environment and the service life of the specific retrofit materials and PV module used, they may need to be replaced. The embodied carbon emissions results of retrofit materials and PV module in this study are as such likely to be underestimated.

During the NHM simulations, fixed regionally specific weather profiles were used to calculate the annual energy consumption of English housing stock without considering the potential impacts of climate change. The monthly climatic conditions are taken from the SAP2012 (Appendix U) regional values and were considered not to change from year to year. With the progress of global warming, the residential energy consumption structure will switch between space heating and space cooling. Leading to higher space cooling load and lower space heating load compared to SAP2012 weather data. We have utilized a constant annual operational energy consumption for 30 years, although the energy saving efficiency of retrofit interventions tends to decrease overtime due to factors such as system degradation and occupant behavior (Department for Business, Energy & Industrial Strategy, 2020b). These would potentially lead to stricter requirements for building retrofit carbon efficiency in order to meet the carbon budgets.

The NHM model also makes in-built assumptions regarding the types of PV modules which result in an electricity generation efficiency of 20%. The efficiency of different modules as deployed may differ in reality, with the efficiencies of monocrystalline, polycrystalline and thin-film PV panels ranging

between 10% to 22% (Sendy, 2020). The embodied carbon emissions of the PV retrofit scenario is likely to be underestimated as we only accounted for PV modules, while a house-integrated PV system have other components, such as invertor and batteries.

For all retrofit scenarios including wall insulation measures, a U-value of 0.1 W/(m²·K) is assumed to have been achieved after retrofit, which is stricter compared to the current maximum requirement of UK building regulations, part L1B of 0.3 W/(m²·K) (HM Government, 2018), and the higher PassivHaus standard requirement of 0.15 W/(m²·K) (Passive House Institute, 2019). In reality, such low U-values may not be feasible to achieve due to their high cost and the space they would physically require to be installed. This would lead to a higher post retrofit U-value and a smaller operational carbon reduction compared to our estimates.

Finally, the carbon budget projection in this study is based on the historical proportions of carbon emissions from residential stock. Depending on the difficulty of carbon reduction in various different sectors, the carbon budget assigned to residential stock might be higher or lower than what is estimated here.

4 Conclusions and Policy Implications

By combining the operational carbon calculated based on the National Household Model and cradle-to-gate embodied carbon of retrofit materials and systems in this study, we have investigated the whole-life carbon performance of retrofit measures in the whole English house stock. The results are compared with carbon targets to support retrofit policy making in achieving net zero emissions without overspending the emissions budget. At our current emissions rate, we have shown England's homes to have produced somewhere over 2200 $MtCO_2e$ by 2050. This leads to surpassing the CCC and Tyndall carbon budget by over 1100 $MtCO_2e$ and over 1800 $MtCO_2e$ respectively under S1, S5 and S6. We have also shown that even under large-scale residential retrofit scenarios, total emissions consistently overshoot the 2050 carbon budget, with the exception of those aggressively deploying heat pumps and thus electrifying heating. Given our implicit assumption that all retrofit

measures have already been implemented at the start of 2021, the following policy recommendations are becoming increasingly more essential in achieving net zero by 2050 within in the residential sector:

1. Rapidly increased heat pump deployment, without the current planned delays until 2028. Given nearly 20 million homes in England alone are eligible for heat-pumps, installing at the pace of current targets, i.e., achieving a deployment rate of only 600 thousand heat pumps annually by 2028, will require an additional 25 years after 2028 before every home is converted.
2. Building envelope retrofit use carefully selected material to bring down the embodied carbon emissions from retrofit, whilst reduce the heating demand for maintaining comfortable indoor environment. Thus leading to less electricity consumption during operation and lower electricity bills for heating when switched to heat pump systems.
3. Current targets for offshore wind are insufficient to achieve net zero carbon by 2050 and need to be made more ambitious. Our estimates suggest deployment of over 11 thousand additional 2 MW wind turbines in the beginning of 2021 is required in tandem with the nationwide installation of heat pumps and double glazing in order to meet Tyndall carbon budget for English house stock.

Measures 1 and 2 need to be deployed across all homes, creating challenges in financing and incentivising what can be disruptive retrofits. Thus, there is a need to increase awareness about the potential energy/carbon/cost savings, from residential retrofit, in addition to the thermal comfort benefits. Attractive financing options will need to be provided to enable all home owners to undertake retrofits given their key role in delivering widespread decarbonisation, and these need to be made available near immediately given the pace of change we have shown is required to meet carbon budgets. Measure 3 is in some respects simpler in that it is not at the mercy of millions of individual

home owners, but the need is no less urgent and requires significant government investment and leadership.

Funding Statement HA, DDT, GB, and TO time was supported by EPSRC Active Building Centre program of work [EP/S016627/1, EP/V012053/1]; DDT time was also supported by EPSRC Multi-Scale, Circular Economic Potential of Non-Residential Building Stock [EP/S029273/1]; XL time was supported by EPSRC UKCRIC-CORONA: City Observatory Research Platform for Innovation and Analytics and Active Building Centre program [EP/R013411/1, EP/V012053/1].

Data Accessibility The data used for the analysis presented in this paper is available at the online supplementary materials for this paper.

Competing Interests The others declare no competing interests.

Author Contributions TO and DDT conceived and designed the analysis, XL and GB collected the data, XL, GB, and H.A. performed the analysis, HA, XL, GB, DDT, and TO wrote the paper.

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